

Graham Alexander Noblit, PhD

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Applied ML leader with 8+ years across industry and research, spanning production search and retrieval, generative modeling, and statistical inference. Manages applied ML scientists and directs cross-functional programs from opportunity discovery and assessment through architecture, evaluation, and deployment.

SKILLS

Machine Learning: Search, Retrieval and Reranking; Generative Modeling (Diffusion, Transformers); Natural Language Processing; Evaluation Design

Statistics: Statistical Inference; Regression (Mixed-effects, Spatial) & Econometrics; Causal Inference & Experimental Design

Engineering: Python, SQL, R, Triton Inference Server, Model Serving, Kubernetes, Docker, AWS

PROFESSIONAL EXPERIENCE

Electronic Arts, Lifestyle Advanced Technology Group

Kirkland, WA

Senior AI Data Scientist and Manager

February 2025 – Present

- Direct technical discovery across an applied ML portfolio, assessing feasibility, defining success criteria, and converting opportunities into execution plans with product, design, and engineering.
- Manage two applied ML scientists, setting priorities, assigning ownership, reviewing system designs and experimental results, and providing technical coaching and delivery oversight.
- Re-architected a Triton-based asset-retrieval service using hierarchical semantic search, patch-level reranking, and open-vocabulary retrieval, reducing end-to-end latency by 50% and increasing mean reciprocal rank by 20%.
- Designed a graph-based, code-aware developer-support system that achieved a 30% first-attempt resolution rate in internal evaluation, reducing support burden and supporting adoption of a shared monetization platform across the organization.
- Own the roadmap and technical direction for a multimodal search system over user-generated content in The Sims, a franchise with more than 50 million lifetime players; define retrieval architecture and evaluation strategy to improve asset discovery and reuse across teams.
- Scoped, architected, and built a secure cloud platform for generative thumbnail workflows, including role-scoped contractor access, isolated Kubernetes workloads, Figma integration, and a diffusion backend; directed a report's LoRA-based style-alignment work.
- Define model and infrastructure strategy for a player-customization platform and lead development of a rectified-flow diffusion system designed to replace an unscalable render-farm workflow and remove a technical blocker to release.

Children's Hospital Los Angeles

Los Angeles, CA

Senior Data Scientist, Team Lead

September 2023 – February 2025

- Served as the principal ML and statistical advisor across clinical and research initiatives, assessing project viability, defining analytical approaches and success criteria, and guiding clinical and research teams through execution.
- Supervised a team of data scientists, reviewing technical approaches, establishing delivery plans, and mentoring team members on machine learning, statistics, and recent advances in AI.

- Raised clinical-note classification F1 from 0.50 to 0.86 by designing language-model-based information extraction and retrieval pipelines incorporating self-supervised learning, semantic search, and retrieval-augmented workflows.
- Reframed a genomics ML initiative after identifying methodological risks in its original design; redefined project scope, data collection, and modeling strategy using Bayesian mixture models and clustering.
- Reduced experimental costs by 10% by developing machine learning and Bayesian regression methods that recovered usable information from otherwise discarded datasets.
- Advised 100+ clinical, research, and administrative stakeholders on practical LLM adoption, implementation considerations, and organizational implications; received the highest audience rating in the history of the institute-wide speaker series.

Schwartz Reisman Institute for Technology and Society; Vector Institute for AI

Toronto, ON

Research Scientist

September 2022 – September 2023

- Defined the core approach for a cross-disciplinary MARL project on norm formation, adapting a cultural-evolution algorithm from doctoral research and coordinating implementation and evaluation with economists and computer scientists.
- Conceived and authored a cultural-evolution paper developing a game-theoretic model of how norms emerge and persist under varying incentives and interaction structures.
- Led weekly planning for a 15-person interdisciplinary AI conference team, developing programming themes and recruiting speakers, and mentored doctoral researchers.

Independent Research Consultant — Part-Time

August 2019 – September 2023

- **DeepMind:** Co-designed a multi-agent reinforcement-learning algorithm for domain-general cooperation using norm-based enforcement.
- **University of Chicago:** Built geospatial datasets and developed mixed-effects and spatial-regression analyses supporting research published in PNAS.
- **Rare:** Designed the research methodology and field data-collection strategy for a behavioral-change initiative focused on fishery sustainability in the Brazilian Amazon.

Harvard University

Cambridge, MA

Doctoral Researcher

August 2016 – May 2022

- Designed and executed empirical studies of human social and economic behavior, integrating climatic, agricultural, economic, geospatial, and survey data and applying causal inference, spatial regression, dimensionality reduction, and game-theoretic optimization across three scientific articles.
- **Head Teaching Fellow, Statistics (2019–2021):** supervised teams of five to nine teaching assistants and taught hypothesis testing, model selection, R programming, and interpretation of statistical results.

EDUCATION

Harvard University, PhD in Human Evolutionary Biology, Cambridge, MA

May 2022

University of Texas at Austin, BA in Anthropology, Minor in Mathematics, Austin, TX

May 2012